Introduction to Robots

Definition of a Robot

- "A reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks" .

Or a simpler version

- An automatic device that performs functions normally ascribed to humans or a machine in the form of a human.

Introduction to Robots

A general-purpose, programmable machine possessing certain anthropomorphic characteristics

- Hazardous work environments
- Repetitive work cycle
- Consistency and accuracy
- Difficult handling task for humans
- Multi shift operations
- Reprogrammable, flexible
- Interfaced to other computer systems

Introduction to Robots

Specifications of robotics

- Axis of motion
- Work stations
- Speed
- Acceleration
- Pay load capacity
- Accuracy
- Repeatability

Introduction to Robots

Historical Evolution:

- **Early Mechanical Automatons**:

Ancient Greeks and Egyptians built mechanical devices resembling humans and animals for entertainment and religious purposes.

- **Industrial Revolution (18th-19th Century):**

Introduction of mechanical machines powered by steam, later electricity, for manufacturing.

- **20th Century Developments:**

Invention of programmable robots for assembly lines (e.g., Unimate in 1961).

- **21st Century:**

Integration of artificial intelligence, sensors, and advanced algorithms.

Emergence of collaborative robots (Cobots) and autonomous systems.

Introduction to Robots

Importance:

- **Enhanced Productivity**: Robots can work continuously without fatigue, increasing output and efficiency.
- **Improved Precision and Accuracy**: Robots excel in tasks requiring high precision, such as surgery and electronics manufacturing.
- **Safety and Risk Mitigation**: Robots perform dangerous tasks, protecting human workers (e.g., bomb disposal, deep-sea exploration).
- **Applications Across Industries**: Found in sectors like healthcare, automotive, logistics, and agriculture.

Basic Structure of Robots

I. Mechanical Components:

1. **Links**: The rigid members or parts of the robot that connect different joints. Serve as the "bones" of the robot, providing structure.

2. **Joints**: Provide motion between the links. Types:

- **Rotational Joints**: Allow rotation.
- **Linear Joints**: Enable straight-line motion.

3. **Actuators**: Devices that convert energy into motion. Types:

- **Electric Actuators**: Motors powered by electricity (most common).
- **Hydraulic Actuators**: Use fluid pressure for heavy lifting.
- **Pneumatic Actuators**: Use compressed air for lighter tasks.

Basic Structure of Robots

II. Electrical Components

1. **Power Supply**: Provides the energy needed to run actuators, sensors, and controllers. Types:

- Batteries (portable robots).
- Electrical power from external sources (industrial robots).

2. **Wiring and Circuitry**:

- Transmits signals and power between components.
- Ensures smooth communication within the robot.

Basic Structure of Robots

III. Control Systems

1. **Controller**: Acts as the "brain" of the robot. Receives input from sensors, processes it, and sends commands to actuators.

2. **Microprocessors and Microcontrollers**:

Process data and execute programs for robot operation.

Modern robots use advanced processors for AI and machine learning.

Basic Structure of Robots

IV. Sensors

Purpose: Detect environmental conditions or the state of the robot. Provide feedback to the controller for decision-making.

Common Types:

Proximity Sensors: Detect objects nearby.

Vision Sensors: Cameras for image processing.

Tactile Sensors: Mimic the sense of touch.

Gyroscopes/Accelerometers: Measure orientation and motion.

Basic Structure of Robots

V. End Effector - The tool or device attached to the robot's manipulator for performing tasks.

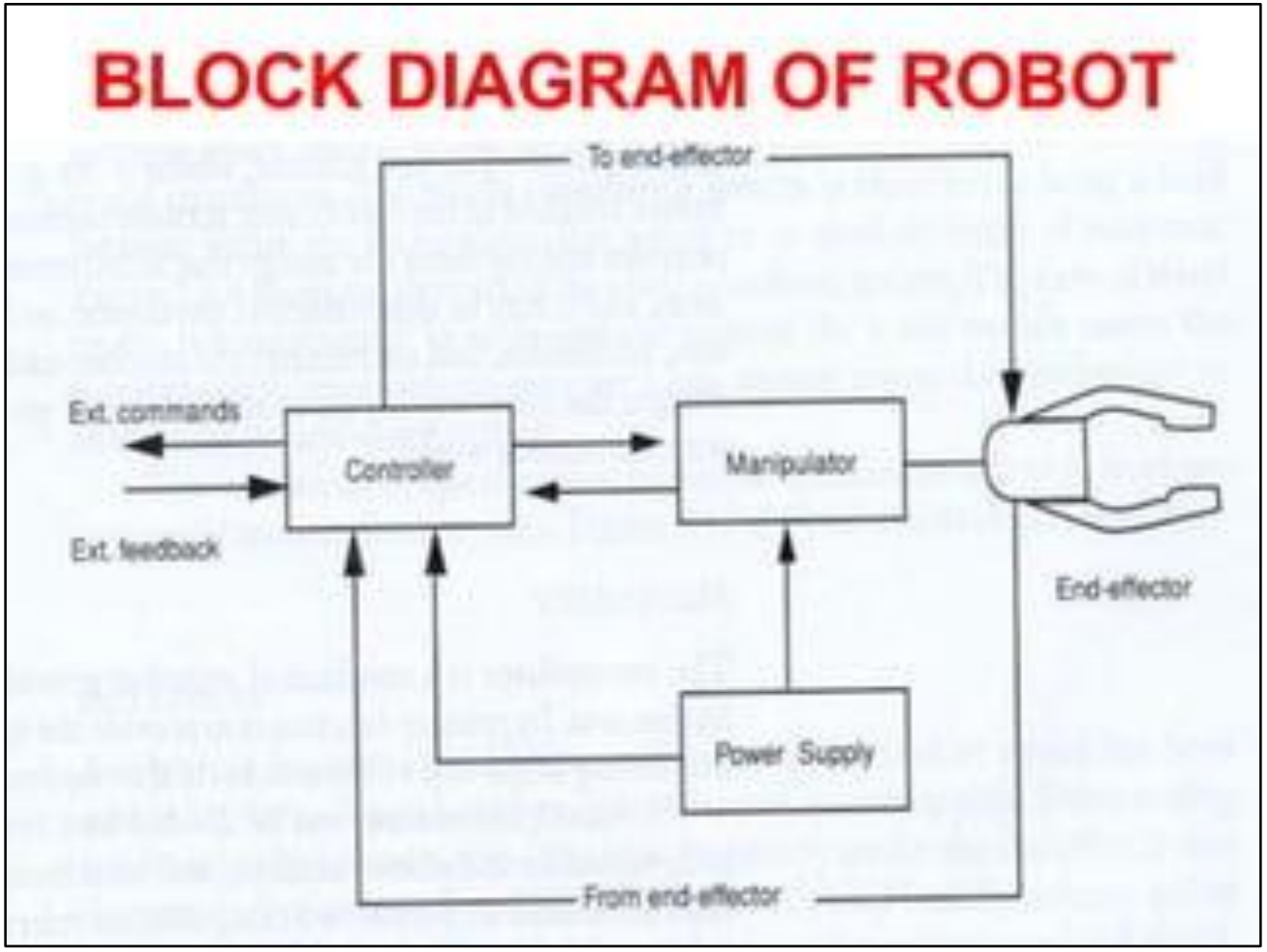
Examples: Grippers, welding torches, painting guns, or cameras.

VI. Mobility Systems (For Mobile Robots)

Wheels or Tracks: Enable movement on flat or rugged surfaces.

Legs: Used in robots designed for walking or climbing.

Basic Structure of Robots



Robot Anatomy

Robot anatomy refers to the physical structure and components that enable a robot to interact with its environment, perform tasks, and carry out actions. The anatomy of a robot includes several key parts, each serving specific functions, similar to how human anatomy enables movement and function.

Robot Anatomy

1. Manipulator

Definition: The "arm" or "limb" of the robot used to interact with objects in its environment. It is typically composed of links and joints.

Function: Similar to a human arm, it helps the robot perform tasks such as lifting, moving, or holding objects.

Components:

- **Links**: Rigid parts that connect joints, similar to bones.
- **Joints**: Allow movement and flexibility. Types include rotational joints (rotate) and prismatic joints (move linearly).

Robot Anatomy

2. End Effector

Definition: The tool or device attached to the end of the manipulator (robot's "hand"). The end effector is used for specific tasks.

Function: Executes the action that the robot was programmed to do, such as picking up, welding, painting, or assembling.

Examples:

- **Grippers:** Used for picking up and holding objects.
- **Welders:** Used for welding parts together.
- **Paint Guns:** For painting objects in industries like automotive manufacturing.
- **Suction Cups:** For handling delicate or fragile items.

Robot Anatomy

3. Mobility Systems (for Mobile Robots)

Definition: Mobility systems provide robots with the ability to move and navigate within their environment. They are critical for mobile robots such as autonomous vehicles and drones.

Types of Mobility:

- **Wheels:** Common in robots designed to move on flat surfaces (e.g., warehouse robots).
- **Tracks:** Used in robots that need to navigate rough or uneven terrain (e.g., military robots or exploration robots).
- **Legs:** Used in walking robots that can navigate complex environments (e.g., Boston Dynamics' robots like Spot).

Robot Anatomy

4. Sensors

Definition: Sensors enable the robot to gather information about its environment or the status of its own components. They provide data that allows the robot to make decisions and adapt its behavior.

Types of Sensors:

- **Proximity Sensors**: Detect objects nearby.
- **Vision Sensors (Cameras):** Provide visual input (images or videos) to recognize objects or navigate.
- **Tactile Sensors**: Mimic the sense of touch, useful for delicate handling.
- **Force/Torque Sensors**: Measure the force applied by the robot on an object or in its movements.
- **Gyroscope/Accelerometers:** Help robots maintain orientation and balance, useful in mobile robots and aerial drones.

Robot Anatomy

5. Controller (The Brain)

Definition: The controller is the central processing unit of the robot. It processes input from the sensors and makes decisions based on pre-programmed algorithms or AI models.

Function: The controller sends instructions to the actuators to carry out movements and actions based on sensor feedback.

Components:

- **Microcontroller or Microprocessor:** Executes programs and processes sensor data.
- **Real-time Operating Systems (RTOS):** Ensures timely processing of sensor inputs and actuator outputs.

Robot Anatomy

6. Power Supply

Definition: The power supply provides the energy needed to run all components of the robot, including sensors, actuators, and controllers.

Types:

- **Battery:** Portable power source for autonomous mobile robots.
- **Electricity:** Powered through cables for industrial robots or robots fixed in one place.
- **Fuel Cells:** Used in some robots for long-term energy supply.

Robot Anatomy

7. Feedback Mechanisms

Definition: Feedback mechanisms provide the robot with information about the success of its actions. Sensors monitor changes and send this data to the controller.

Types of Feedback:

- **Positional Feedback:** Ensures the robot is in the right location or orientation.
- **Force Feedback:** Indicates the force exerted on an object.
- **Visual Feedback:** Used by vision systems to recognize objects or map environments.

Robot Anatomy

How Robot Anatomy Works Together

- Sensors gather data about the environment.
- Controller processes the sensor data and decides on the next action.
- Actuators (such as motors) carry out the physical movements (e.g., turning wheels, moving joints).
- Manipulator and End Effector perform the intended tasks (e.g., lifting, welding, painting).
- Feedback from sensors informs the controller about the success or need for adjustment in the robot’s actions.

Classification of Robots

- By Movement

I. **Fixed Robots**:

Stationary robots with a fixed base.

Examples: Industrial robotic arms in assembly lines.

Applications: Welding, painting, and material handling.

II. **Mobile Robots:**

Robots that can move in their environment.

Types:

- **Wheeled Robots**: Common in indoor environments (e.g., warehouse robots).
- **Tracked Robots**: Used in rugged terrains (e.g., bomb disposal robots).

Examples: Autonomous vehicles, robotic vacuum cleaners.

Classification of Robots

- By Movement

III. Aerial Robots:

Robots capable of flight.

Examples: Drones used for surveillance, delivery, or photography.

Applications: Military reconnaissance, package delivery, disaster management.

IV. Underwater Robots:

Robots designed to operate underwater.

Examples: Remotely Operated Vehicles (ROVs) for deep-sea exploration.

Applications: Oceanography, underwater pipeline inspections.

Classification of Robots

- **By Generations**

I. First-Generation Robots:

Manual robots operated by humans.
Characteristics: No autonomy or sensors.
Examples: Basic robotic arms for assembly tasks.

II. Second-Generation Robots:

Pre-programmed robots with limited adaptability.
Characteristics: No real-time feedback.
Examples: Early CNC machines.

Classification of Robots

III. Third-Generation Robots:

Robots with integrated sensors for feedback.

Characteristics: Capable of adapting to changing environments.

Examples: Robots with proximity and vision sensors.

IV. Fourth-Generation Robots:

AI-based robots with decision-making abilities.

Characteristics: Real-time learning and adaptation.

Examples: Autonomous cars, collaborative robots (Cobots).

V. Fifth-Generation Robots:

Highly autonomous robots with advanced AI.

Characteristics: Ability to collaborate with humans and learn from experience.

Examples: Advanced humanoid robots, social robots.

Classification of Robots

- **By Application**

1. **Industrial Robots:**

Designed for manufacturing and industrial applications.
Examples: Assembly line robots, welding robots.
Applications: Automotive industry, electronics manufacturing.

2. **Service Robots:**

Assist humans in non-industrial environments.
Examples:

- Domestic Robots: Robotic vacuum cleaners, lawn mowers.
- Healthcare Robots: Surgical robots, patient care robots.

Applications: Household chores, healthcare, education.

Classification of Robots

- **By Application**

3. Military Robots:

Used for defense and military purposes.

Examples: Drones, bomb disposal robots, reconnaissance robots.

Applications: Surveillance, combat, and rescue missions.

4. Exploration Robots:

Used for space, deep-sea, or inaccessible environments.

Examples: Mars rovers, deep-sea ROVs.

Applications: Space exploration, oceanography.

Classification of Robots

- **By Control Mechanism**

1. **Manual Control:** Operated directly by a human.

Examples: Remote-controlled drones.

2. **Automatic Control:** Pre-programmed robots with no real-time feedback.

Examples: Robotic arms in fixed assembly lines.

3. **Semi-Autonomous Robots:** Perform tasks with some level of human intervention.

Examples: Drones with autopilot features.

4. **Fully Autonomous Robots:** Operate without human intervention, using AI and sensors for decision-making.

Examples: Self-driving cars, warehouse robots.

Classification of Robots

Criteria	Types	Examples
Workspace	Cartesian, Cylindrical, Spherical, Articulated, SCARA	Assembly robots, CNC machines
Interaction with Humans	Collaborative, Autonomous, Humanoid	Cobots, social robots

Fundamentals about Robot Technology

I. Sensors

Definition: Devices that detect environmental or internal conditions and provide data to the robot's control system.

Types of Sensors:

- **Proximity Sensors:** Detect nearby objects without physical contact. Ex: Ultrasonic sensors in autonomous vehicles.
- **Vision Sensors (Cameras):** Capture images or video for object recognition and navigation. Example: Cameras used in warehouse robots.
- **Tactile Sensors:** Mimic the sense of touch to measure pressure or texture. Example: Used in robotic grippers for handling delicate objects.
- **Force/Torque Sensors:** Measure force and torque during operations. Example: Welding robots for precision control.
- **Gyroscope/Accelerometers:** Measure orientation, position, and motion. Example: Balance control in bipedal robots.

Fundamentals about Robot Technology

II. Actuators

Definition: Devices that convert energy into mechanical motion.

Types of Actuators:

- **Electric Actuators:** Motors powered by electricity; commonly used due to precision and efficiency. Example: DC motors in robotic arms.
- **Hydraulic Actuators:** Use pressurized fluids for heavy-duty tasks. Example: Excavation robots.
- **Pneumatic Actuators:** Use compressed air for lightweight and rapid operations. Example: Assembly line robots.

Importance: Actuators provide the movement required to perform tasks, such as picking up objects or navigating environments.

Fundamentals about Robot Technology

III. Control Systems

Definition: Systems that manage and coordinate the operation of sensors, actuators, and other components.

Types of Control Systems:

- **Open-Loop Control:** No feedback mechanism; actions are pre-programmed. Example: Conveyor belts with fixed timing.
- **Closed-Loop Control:** Uses feedback from sensors to adjust actions in real-time. Example: Temperature control in welding robots.

Key Functions:

- Process data from sensors.
- Decide actions based on programming or AI algorithms.
- Send signals to actuators for motion or task execution.

Fundamentals about Robot Technology

IV. Power Systems

Definition: Provide the energy required for all robot components.

Types of Power Sources:

- **Battery:** Used in mobile robots for portability. Example: Lithium-ion batteries in drones.
- **Electricity:** Wired power source for stationary robots. Example: Robotic arms in factories.
- **Fuel Cells:** Used for long-duration tasks requiring higher energy density. Example: Underwater robots.

Importance: Ensures uninterrupted operation of the robot's systems.

Fundamentals about Robot Technology

V. Programming and Algorithms

Definition: The set of instructions that control a robot's actions.

Programming Approaches:

1. **Manual Programming**: Operators input instructions directly. Example: Robot arm programmed for specific welding paths.
2. **Adaptive Programming**: Robots learn from sensor data and adapt actions. Example: Self-driving cars adjusting to traffic conditions.

Use of Algorithms:

- **Pathfinding:** Determine the shortest route in a workspace.
- **Object Recognition:** Identify objects using vision systems.
- **AI and Machine Learning:** Allow robots to make decisions and improve over time.

Fundamentals about Robot Technology

V. Feedback Systems

Definition: Mechanisms that allow robots to adjust actions based on real-time data.

Components of Feedback:

- **Sensors:** Provide data on performance and environment.
- **Controller:** Processes feedback and recalibrates actions.
- **Output Adjustment:** Ensures precise and accurate task execution.

Importance: Critical for tasks requiring high precision, such as robotic surgery.

Fundamentals about Robot Technology

Applications of Robot Technology

- **Industrial Manufacturing**: Assembly, painting, welding, and material handling.
- **Healthcare**: Robotic-assisted surgeries, rehabilitation devices.
- **Service Robots**: Cleaning, delivery, and customer service robots.
- **Exploration**: Space rovers, deep-sea robots for inaccessible terrains.
- **Military:** Drones, bomb disposal robots, and surveillance systems.

Fundamentals about Robot Technology

- **Real-Time Sensing for Adaptation**: Sensors gather environmental data, enabling robots to adapt autonomously to changing conditions and execute tasks with fine adjustments.
- **Precise Motion via Actuators**: Actuators, such as electric and hydraulic systems, provide controlled movements essential for high-accuracy operations like assembly or surgery.
- **Intelligent Control Systems**: Closed-loop control systems process feedback to make real-time corrections, ensuring tasks are completed with minimal error.
- **Feedback Mechanisms for Accuracy**: Continuous monitoring through feedback loops allows robots to self-correct actions, enhancing precision in repetitive or delicate tasks.
- **Advanced Programming and AI Algorithms**: Algorithms enable robots to execute complex operations autonomously, while machine learning refines precision over time through adaptive behavior.

Basic Configurations of Robots

- **Cartesian Robots**

Description: Operate in a rectangular workspace with linear motion along three perpendicular axes (X, Y, Z).

Characteristics:

- Simple design and easy to program.
- High accuracy and repeatability in operations.

Applications: Pick-and-place tasks, CNC machining, 3D printing.

Advantages:

- Easy to maintain and program.
- High stiffness for precise tasks.

Disadvantages:

- Limited range of motion compared to other configurations.

Basic Configurations of Robots

- **Cylindrical Robots**

Description: Combine linear motion along one axis with rotational motion around a base.

Characteristics:

- Cylindrical work envelope.
- Suitable for operations requiring rotational access.

Applications: Assembly tasks, material handling, and welding.

Advantages:

- Compact and flexible design for specific tasks.

Disadvantages:

- Limited reach compared to Cartesian robots.

Basic Configurations of Robots

- **Spherical Robots**

Description: Feature a pivoting arm and rotary joints that create a spherical workspace.

Characteristics:

- Combine rotational and linear movements.

Applications:

Inspection, surveillance, and arc welding.

Advantages:

- Wide reach within a spherical range.

Disadvantages:

- Complex design and programming.

Basic Configurations of Robots

- **Articulated Robots**

Description: Multi-jointed robots with rotary joints, resembling a human arm.

Characteristics:

- Highly flexible and versatile.
- Work envelope depends on the number of joints (Degrees of Freedom).

Applications: Painting, welding, assembly, and material handling.

Advantages:

- Excellent flexibility for complex tasks.

Disadvantages:

- Expensive and challenging to program.

Basic Configurations of Robots

- **SCARA Robots**

Description: Designed for high-speed assembly and precision tasks within a limited workspace.

Characteristics: Horizontal motion for compliance in specific directions.

Applications: Electronics assembly, palletizing, and machine loading.

Advantages:

- High speed and precision.
- Compact design for limited spaces.

Disadvantages:

- Restricted to specific tasks and small workspaces.

Basic Configurations of Robots

Configuration	Motion Type	Work Envelope	Common Applications	Advantages	Disadvantages
Cartesian	Linear	Rectangular	CNC machining, 3D printing	High precision, simple design	Limited range of motion
Cylindrical	Linear + Rotary	Cylindrical	Assembly, welding	Compact, flexible	Limited reach
Spherical	Rotary + Linear	Spherical	Surveillance, inspection	Wide reach	Complex design
Articulated	Multi-rotary	Various	Welding, painting	Highly flexible	Expensive, complex to program
SCARA	Linear + Rotary	Horizontal plane	Electronics assembly	High speed and precision	Limited workspace

Applications and Trends in Robotics

I. Industrial Applications

Manufacturing and Assembly: Robots automate repetitive tasks such as welding, painting, and assembling components. Example: Robotic arms on automotive production lines.

Material Handling: Robots transport, sort, and organize materials in warehouses. Example: Automated Guided Vehicles (AGVs) in e-commerce warehouses.

Quality Control: Vision-enabled robots inspect products for defects. Example: Robots in electronics manufacturing.

Applications and Trends in Robotics

II. Healthcare

Surgical Robots: Assist surgeons in minimally invasive surgeries with high precision. Example: da Vinci Surgical System.

Rehabilitation: Robots aid in physical therapy for patients with mobility impairments. Example: Exoskeletons for walking assistance.

Elderly Care: Companion robots provide emotional support and assistance in daily tasks. Example: Robots like Paro for dementia patients.

Applications and Trends in Robotics

III. Service Robotics

Domestic Robots: Perform household tasks such as cleaning and lawn mowing. Example: Robotic vacuum cleaners like Roomba.

Customer Service: Robots assist in retail and hospitality sectors. Example: Pepper robot for customer interaction.

Delivery Robots: Autonomous robots deliver packages or food. Example: Starship delivery robots.

Applications and Trends in Robotics

IV. Military and Defense

Surveillance: Drones monitor borders and gather intelligence. Example: UAVs (Unmanned Aerial Vehicles).

Bomb Disposal: Robots safely neutralize explosive devices. Example: PackBot by iRobot.

Applications and Trends in Robotics

V. Exploration

Space Exploration: Robots explore planets and moons, gathering data. Example: NASA’s Mars rovers (Perseverance, Curiosity).

Underwater Exploration: Robots inspect underwater infrastructure or study marine ecosystems. Example: Remotely Operated Vehicles (ROVs).

Applications and Trends in Robotics

V. Agriculture

Crop Harvesting: Robots automate picking and harvesting. Example: Robots with vision sensors identify ripe fruits.

Weed Control: Autonomous machines remove weeds with precision. Example: Robots like the ecoRobotix.

Applications and Trends in Robotics

1. **Integration with Artificial Intelligence (AI):** Robots are becoming smarter with AI, enabling complex decision-making and autonomous task execution.

2. **Collaborative Robots (Cobots):** Designed to safely work alongside humans, Cobots are increasingly used in manufacturing and logistics.

3. **Industry 4.0 and IoT Integration:** Robots in smart factories collect real-time data, enabling predictive maintenance and process optimization.

4. **Advancements in Autonomous Mobility:** Self-driving cars, delivery robots, and drones are improving navigation in dynamic environments.

5. **Soft Robotics:** Robots made of flexible materials are being developed for delicate and adaptive tasks, such as handling fragile items.

Applications and Trends in Robotics

6. Swarm Robotics: Groups of small robots work together, inspired by the collective behavior of insects, for tasks like search and rescue.

7. Robotics in Healthcare: Applications include surgical robots, rehabilitation aids, and robots for elder care and companionship.

8. Focus on Human-Robot Interaction: Improvements in voice, gesture, and facial recognition enable robots to interact more intuitively with humans.

9. Eco-Friendly and Sustainable Robotics: Robots are being designed with renewable energy sources and biodegradable materials to reduce environmental impact.

10. 3D Printing and Robotics Fusion: Robots equipped with 3D printing capabilities are used for rapid prototyping and large-scale construction projects.